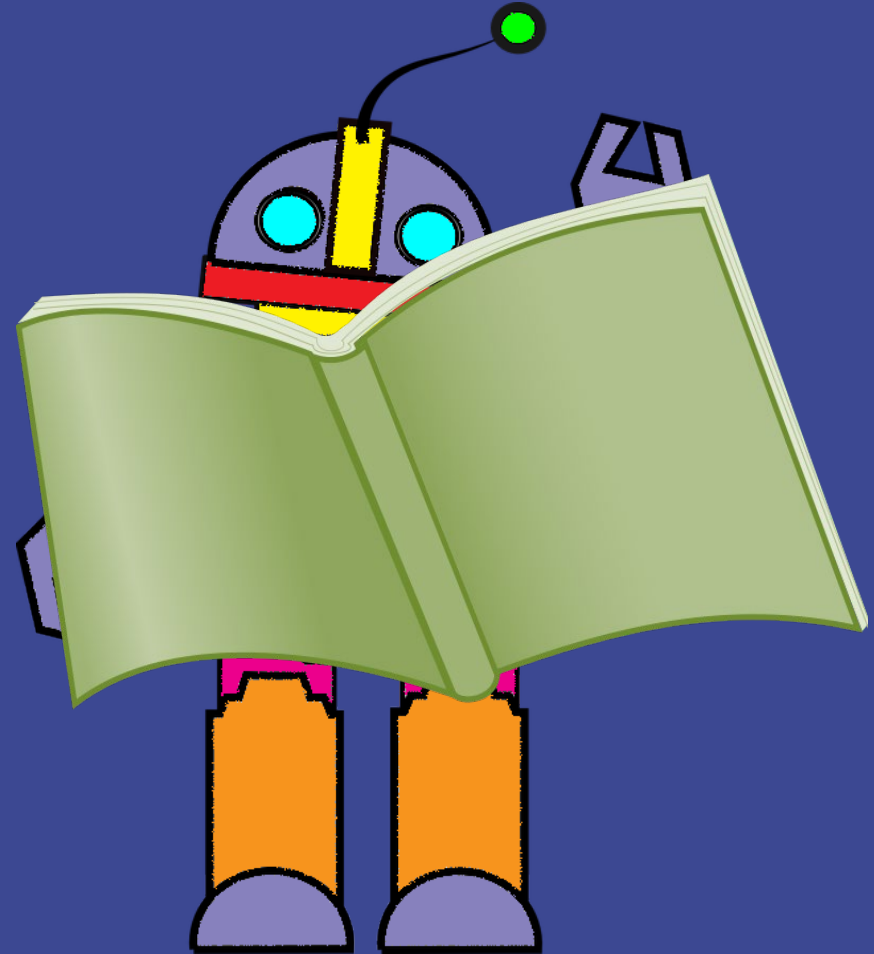


CS 156

Intro to AI

Review



Today

- ▶ Midterm Logistics
- ▶ Review:
 - Tour de AI: Search Algorithms Review
 - Constraint Satisfaction Problems
 - Adversarial Search

Midterm

- ▶ In class, on Thursday
- ▶ On Canvas with LockDown Browser
- ▶ 70 minutes
- ▶ 30 points
- ▶ About 9-11 questions, some with multiple parts
- ▶ Cheat sheet – handwritten
- ▶ Scratch paper (I will provide)

Midterm

- ▶ Intelligent Agents
- ▶ Uninformed Search: DFS, BFS and UCS
- ▶ Informed Search: Greedy and A^* , heuristics
- ▶ Local Search
- ▶ Constraint Satisfaction Problems
- ▶ Adversarial Search, Expectimax, Multi-Agent Utilities

Study Tips

- ▶ Review lectures and prepare/update cheat sheet
- ▶ Retake iClicker quizzes
- ▶ Retake questions of the week
- ▶ Review homework assignments

Exam Tips

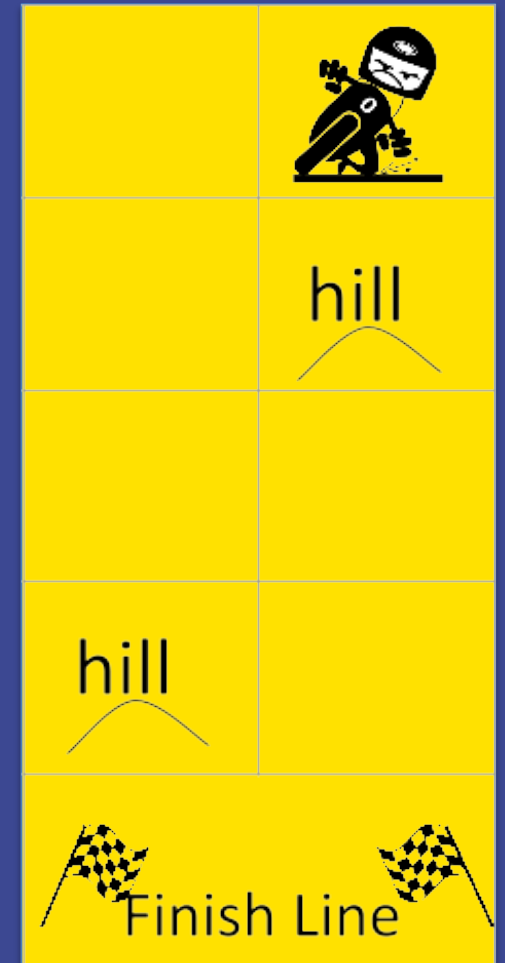
- ▶ Please read the questions carefully
- ▶ Make sure you answer the question that was asked – not something close

Search Algorithms Questions

- ▶ Nodes expanded in the order they were expanded?
- ▶ Solution (path)?
- ▶ Cost?
- ▶ Heuristics:
 - Admissible?
 - Consistent?
- ▶ Local Search

Tour de AI

- ▶ Our task is once again to help our agent find the optimal path to a goal, and today the goal is the finish line.
- ▶ Our agent is riding a bike and must exert more effort when riding up a hill (cost 8).
- ▶ Going down the hill is the easiest (and most fun) with a cost of 1.
- ▶ Biking on a flat terrain has an associated cost of 2 (to go from one grid location to an adjacent one.)
- ▶ Our agent can only travel South ↓ , East → or West ← .



Tour de AI

Start State:

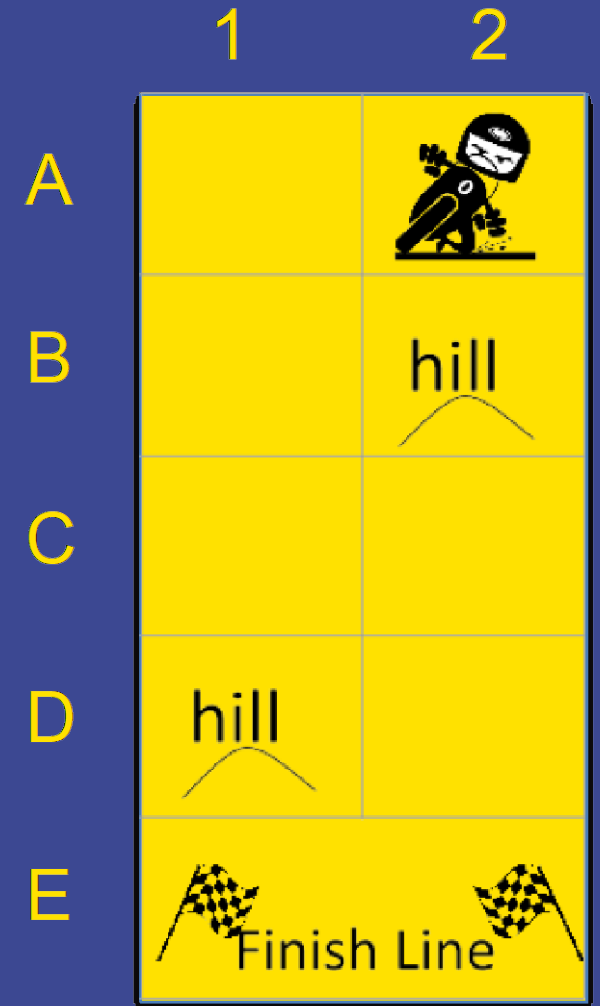
- A2

Goal States:

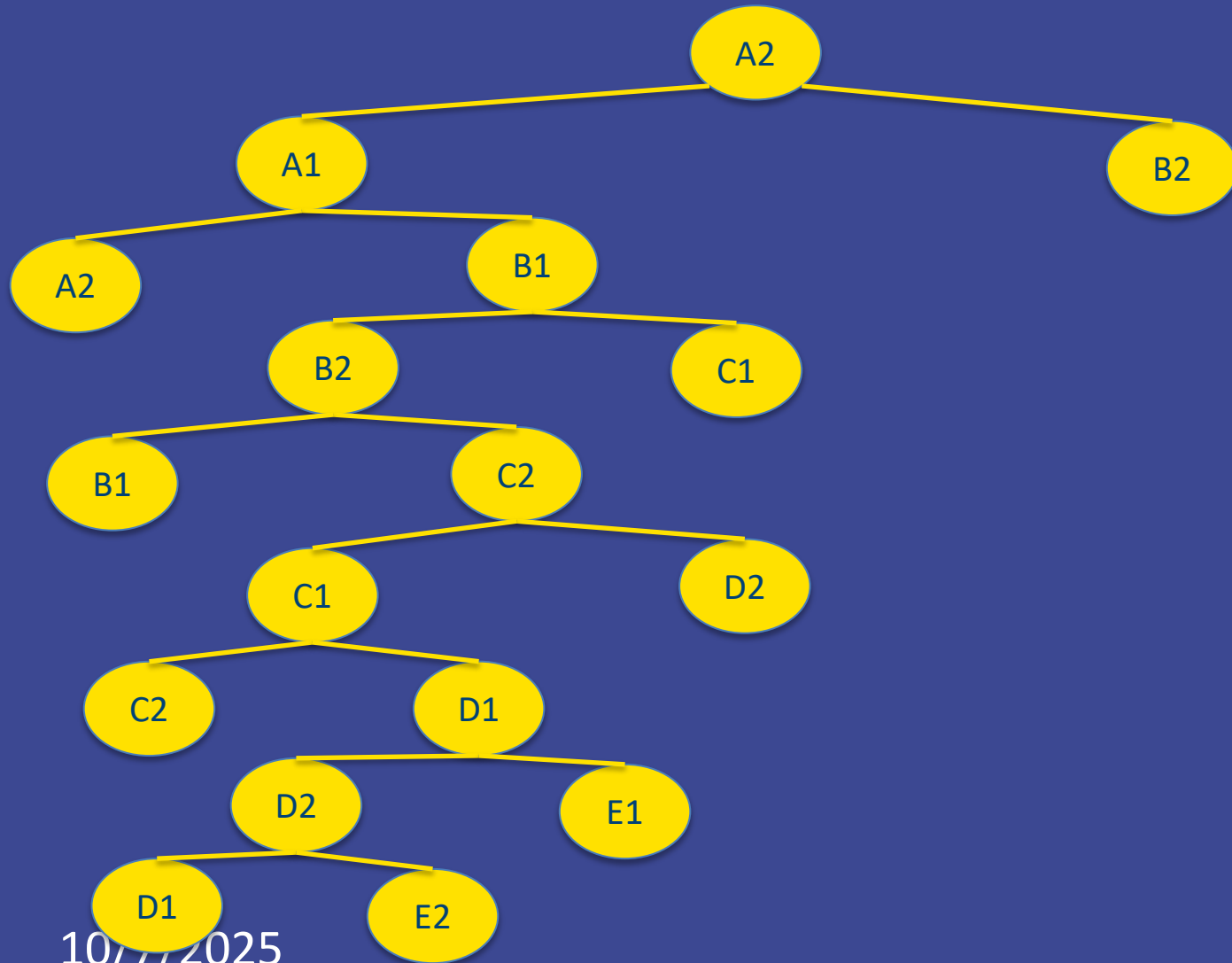
- E1, E2

Costs:

- A2 -> A1: 2
- A2 -> B2: 8
- B2 -> C2: 1
- B2 -> B1: 1...

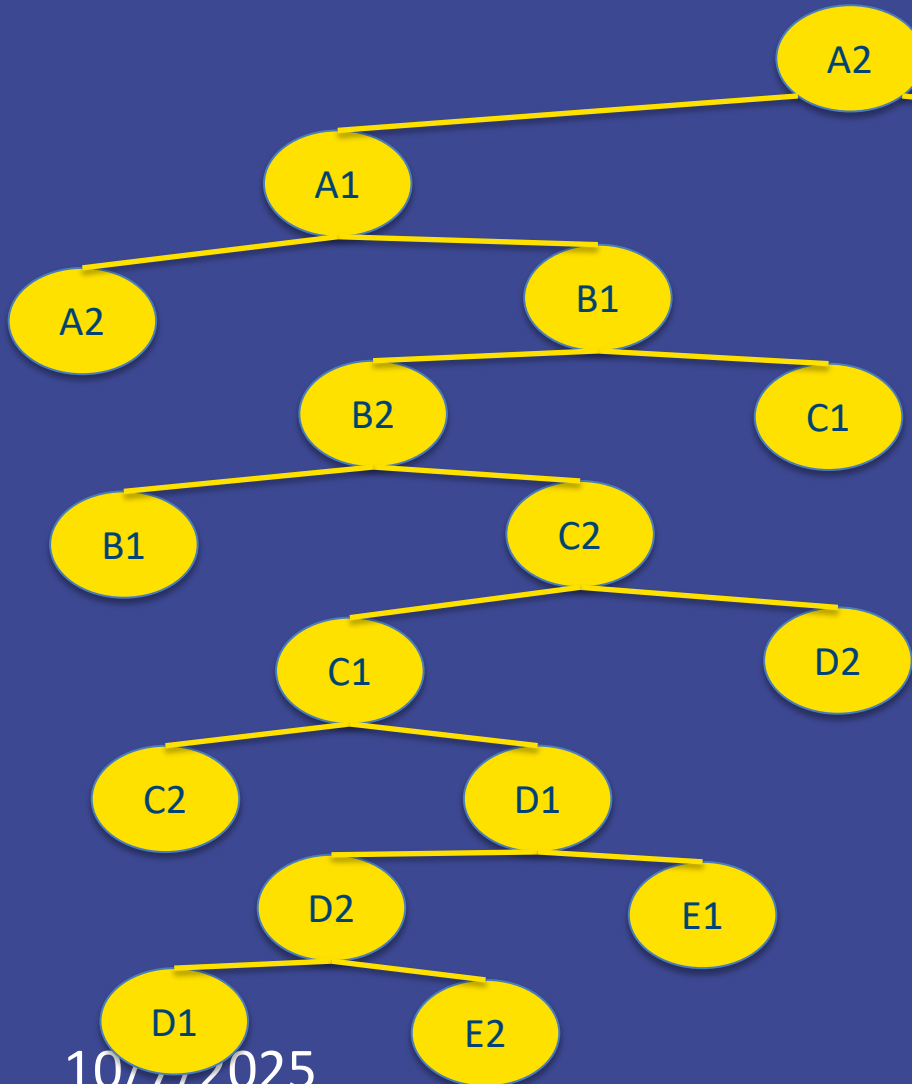


Tour de AI with DFS



	1	2
A		
B		hill 
C		
D	hill 	
E	 Finish Line 	

Tour de AI with DFS



DFS Solution?

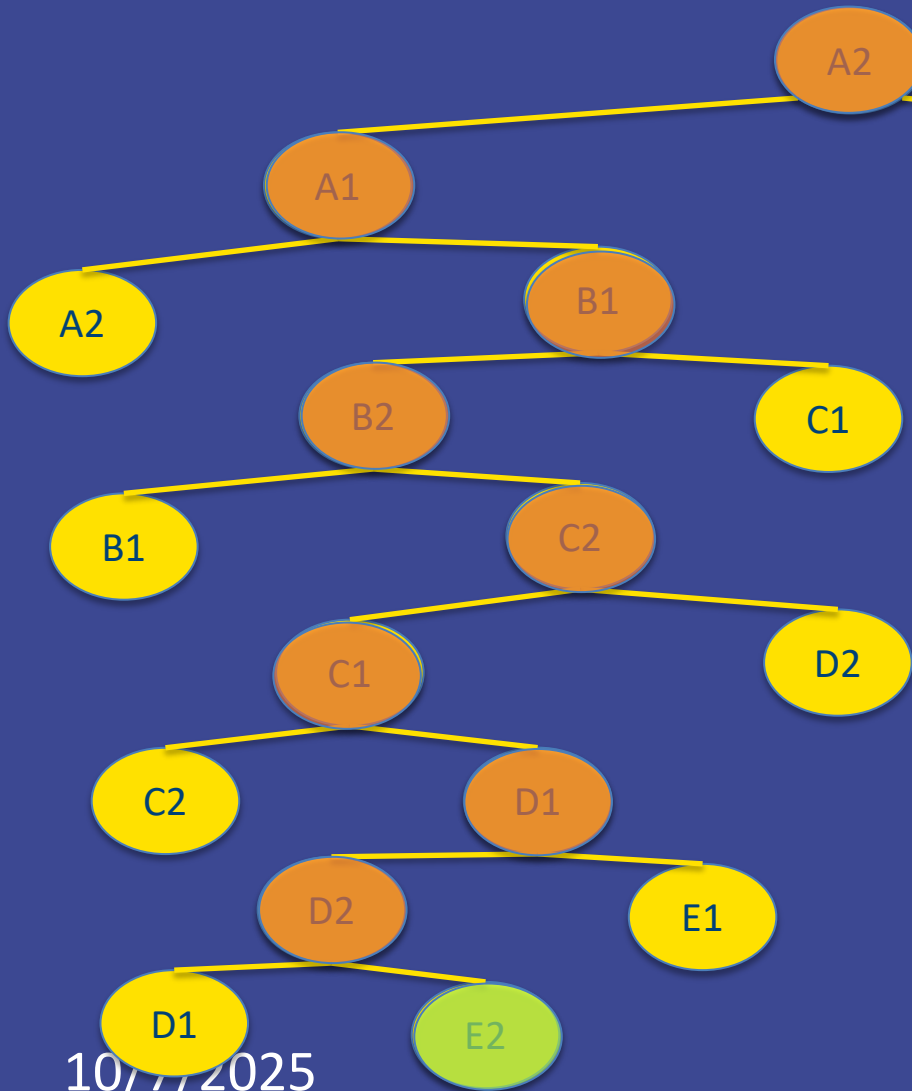
A2A1B1B2C2C1D1D2E2

Cost?

26

	1	2
A		
B		hill 
C		
D	hill 	
E	 Finish Line 	

Tour de AI with DFS

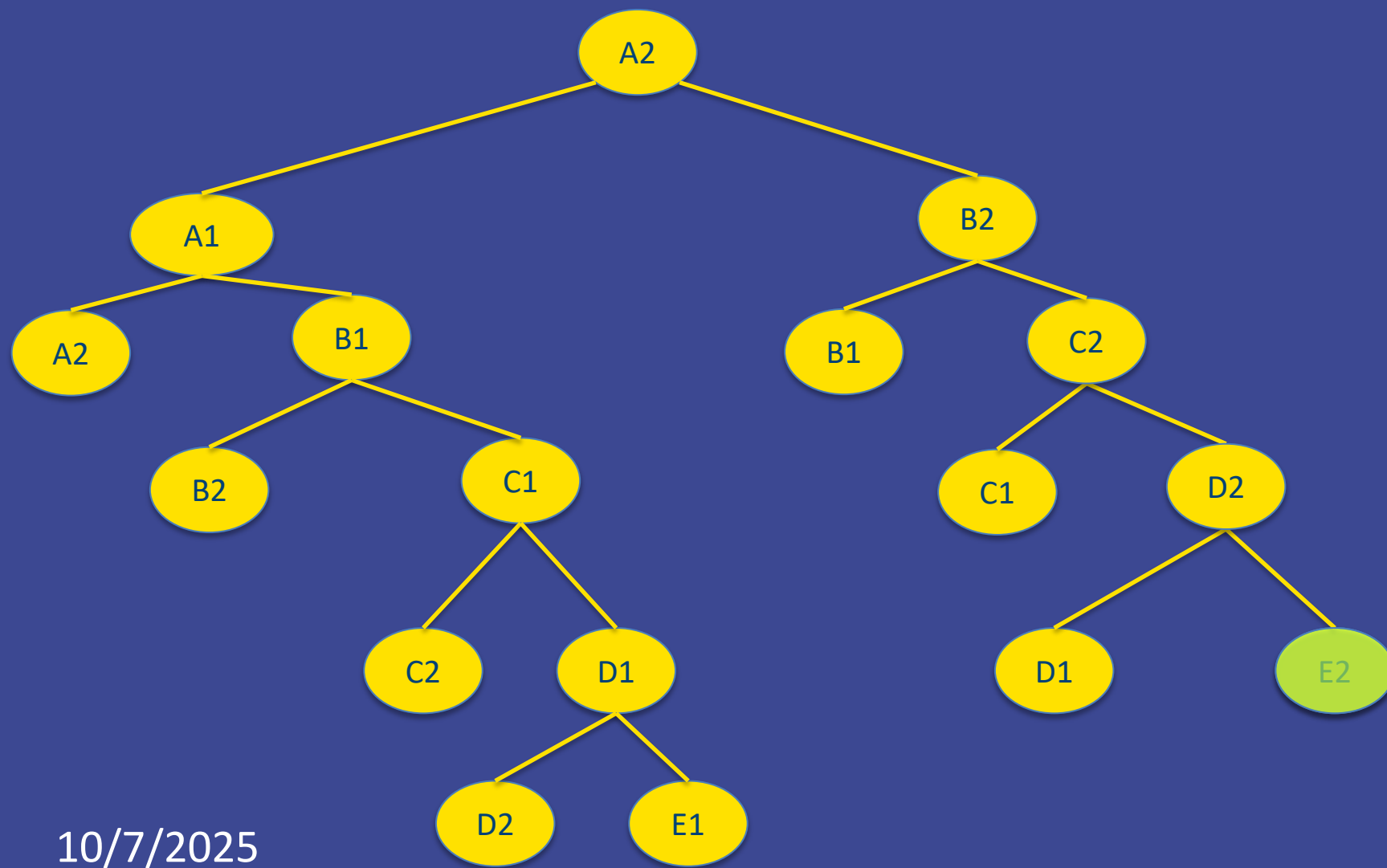


Nodes expanded?

A2
 A2A1
 A2A1B1
 A2A1B1B2
 A2A1B1B2C2
 A2A1B1B2C2C1
 A2A1B1B2C2C1D1
 A2A1B1B2C2C1D1D2

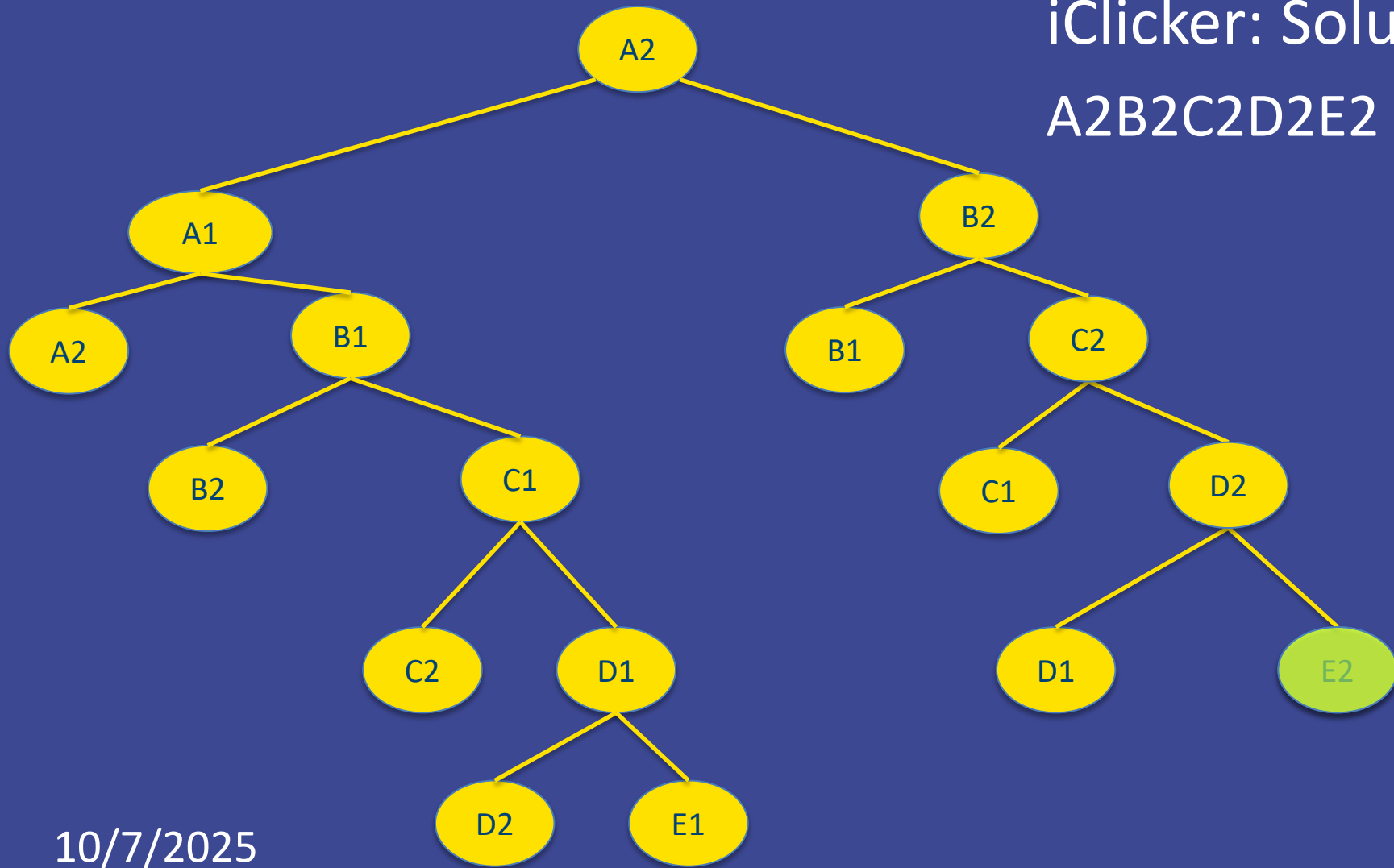
	1	2
A		
B		hill 
C		
D	hill 	
E	 Finish Line 	

Tour de AI with BFS



	1	2
A		
B		hill 
C		
D	hill 	
E	 Finish Line 	

Tour de AI with BFS

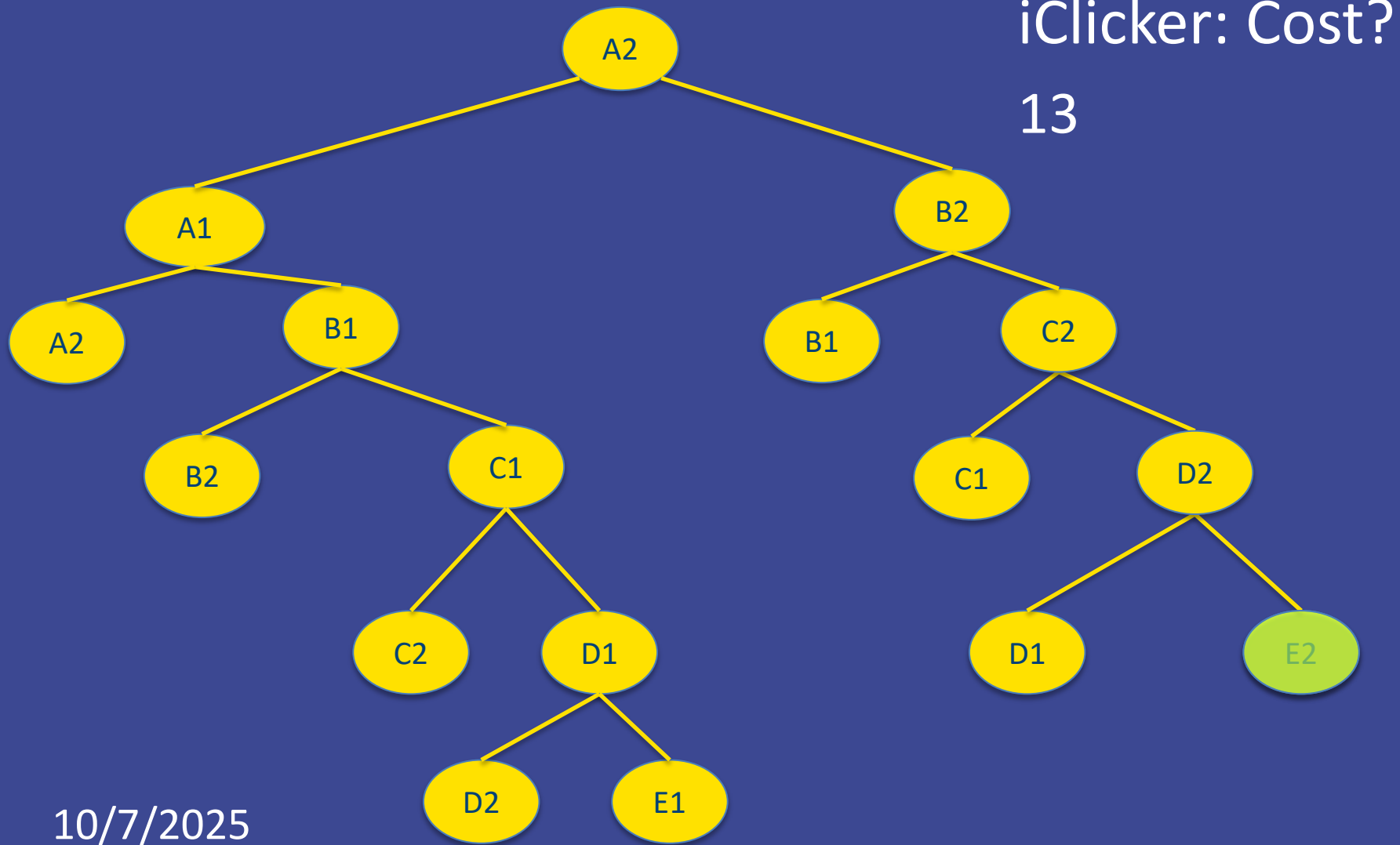


iClicker: Solution?

A2B2C2D2E2

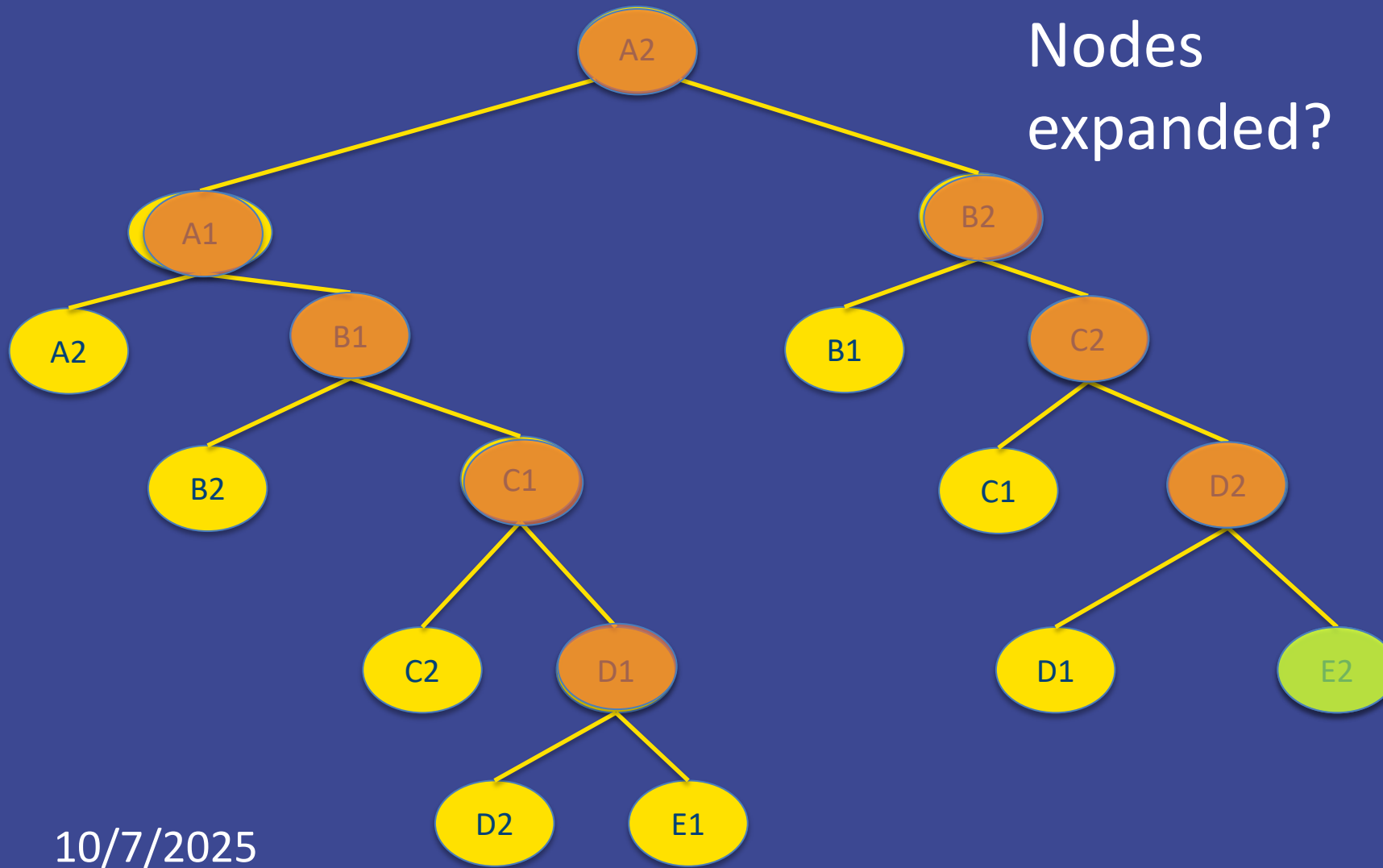
	1	2
A		
B		hill 
C		
D	hill 	
E	 Finish Line 	

Tour de AI with BFS



	1	2
A		
B		hill 
C		
D	hill 	
E	 Finish Line 	
		15

Tour de AI with BFS



A2
A2A1
A2B2
A2A1B1
A2B2C2
A2A1B1C1
A2B2C2D2
A2A1B1C1D1

Heuristics

- ▶ A heuristic is an estimate of the cost from a given state to the goal.
- ▶ We do not add heuristics of multiple states.
- ▶ A heuristic is admissible iff for all states X:
 $h(X) \leq \text{lowest actual cost from X to goal}$
- ▶ A heuristic is consistent iff for all arcs XY:
 $h(X) - h(Y) \leq \text{cost of arc XY}$

Tour de AI with Heuristic

► iClicker: Admissible?

A. Yes

B. No

	1	2
A	8	8 
B	6	6 hill 
C	4	4
D	2 hill 	2
E	 Finish Line 	

Tour de AI with Heuristic

► Admissible?

A. Yes

B. No

	1	2
A	4	4 
B	3	3 hill 
C	2	2
D	1 hill 	1
E	 Finish Line 	

Tour de AI with Greedy

Nodes Expanded?

- ▶ A2
- ▶ A2B2
- ▶ A2B2C2
- ▶ A2B2C2D2

Solution?

A2B2C2D2E2

Cost?

13

	1	2
A	4	4 
B	3	3 hill 
C	2	2
D	1 hill 	1
E	 Finish Line 	

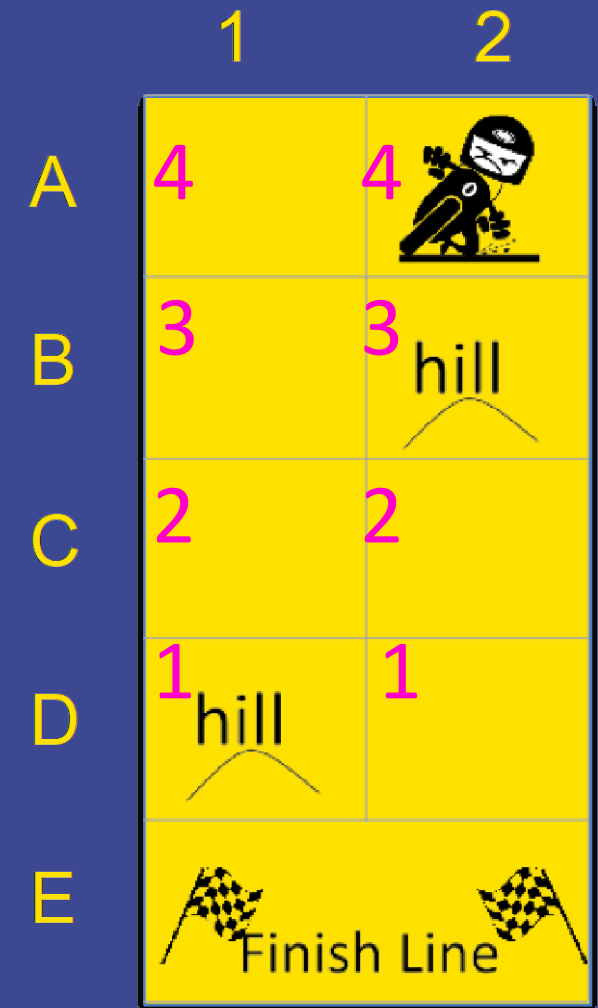
Tour de AI with A*

Node	g	h	f	
A2	0	4	4	Closed set A2 A1 B1
A2A1	2	4	6	
A2B2	8	3	11	
A2A1B1	4	3	7	
A2A1A2	4	4	8	
A2A1B1B2	12	3	15	
A2A1B1C1	6	2	8	

	1	2
A	4	4 
B	3	3 hill 
C	2	2
D	1 hill 	1
E	 Finish Line 	

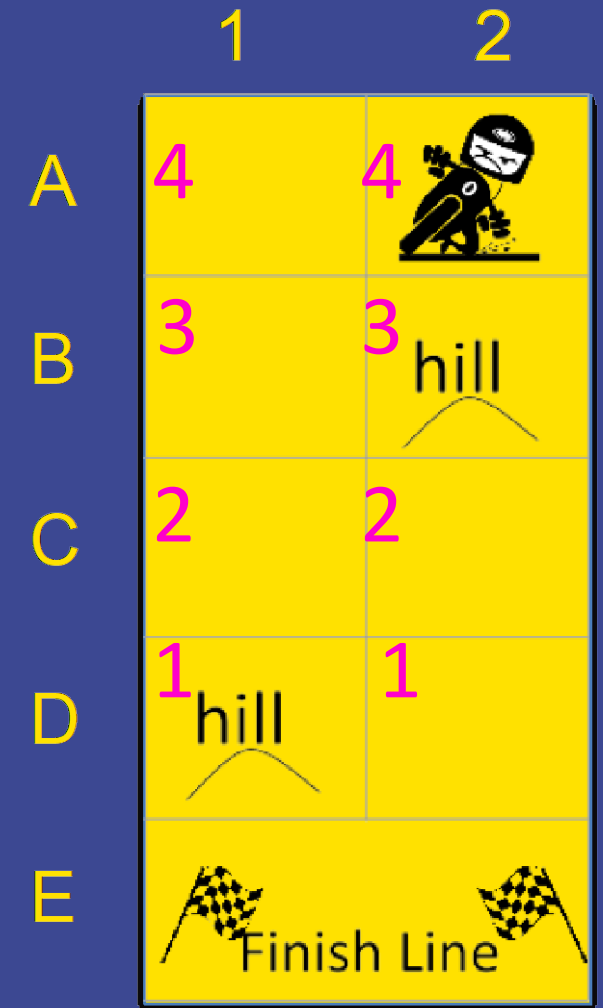
Tour de AI with A*

Node	g	h	f	Closed set
A2B2	8	3	11	A2
A2A1A2	4	4	8	A1
A2A1B1B2	12	3	15	B1 C1
A2A1B1C1	6	2	8	
A2A1B1C1C2	8	2	10	
A2A1B1C1D1	14	1	15	



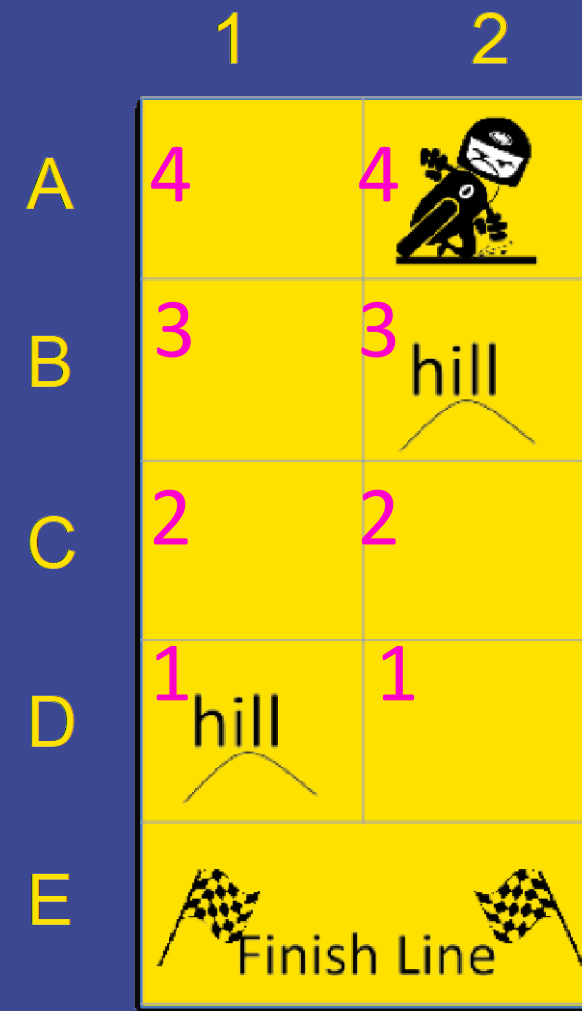
Tour de AI with A*

Node	g	h	f	Closed set
A2B2	8	3	11	A2
A2A1B1B2	12	3	15	A1
A2A1B1C1C2	8	2	10	B1 C1
A2A1B1C1D1	14	1	15	C2
A2A1B1C1C2C1	10	2	12	
A2A1B1C1C2D2	10	1	11	



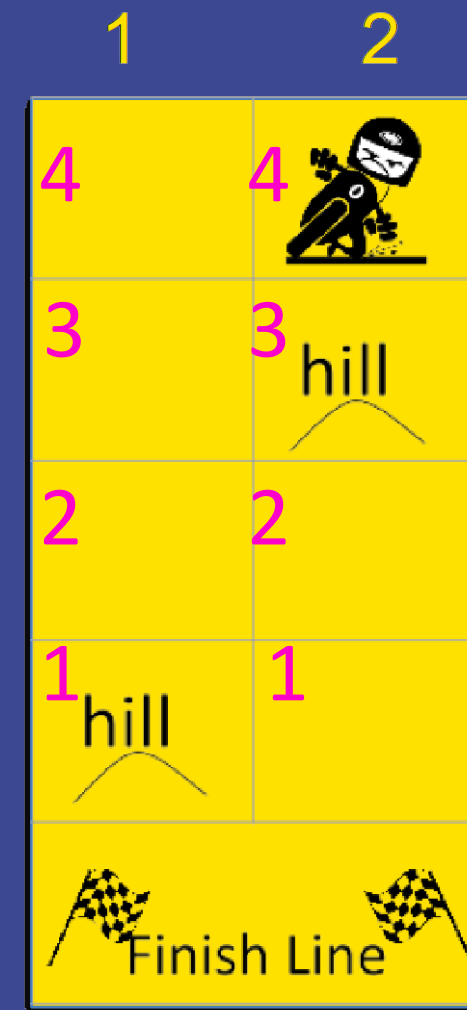
Tour de AI with A*

Node	g	h	f	Closed set
A2B2	8	3	11	A2
A2A1B1B2	12	3	15	A1
A2A1B1C1D1	14	1	15	B1
A2A1B1C1C2C1	10	2	12	C1
A2A1B1C1C2D2	10	1	11	C2
A2B2C2	9	2	11	B2
A2B2B1	9	3	12	



Tour de AI with A*

Node	g	h	f	Closed set	
A2A1B1B2	12	3	15	A2	A
A2A1B1C1D1	14	1	15	A1	
A2A1B1C1C2C1	10	2	12	B1	B
A2A1B1C1C2D2	10	1	11	C1	
A2B2B1	9	3	12	C2	C
				B2	
				D2	D
A2A1B1C1C2D2D1	18	1	19		
A2A1B1C1C2D2E2	12	0	12		E

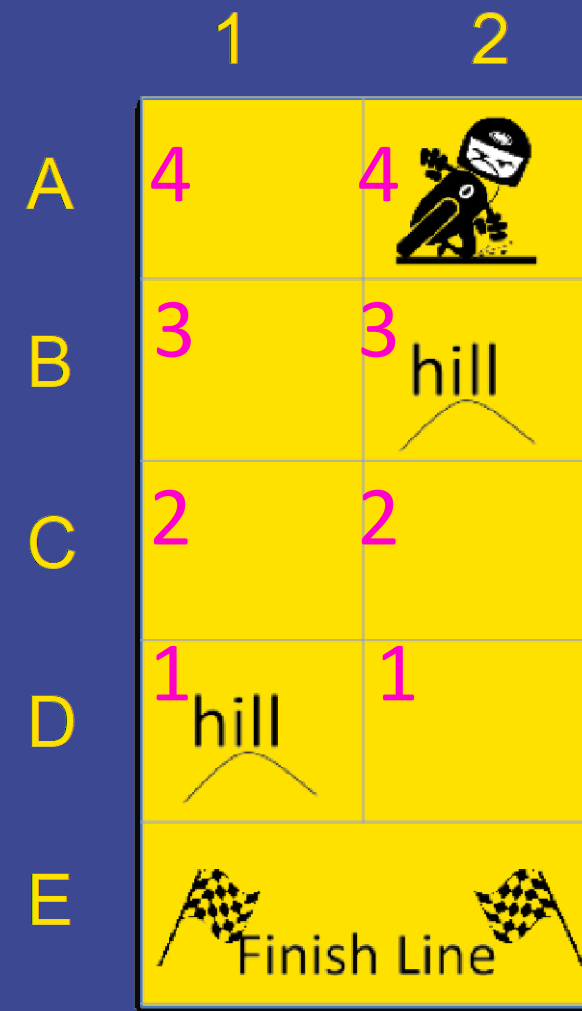


Tour de AI with A*

Node	g	h	f	Closed set
A2A1B1B2	12	3	15	A2
A2A1B1C1D1	14	1	15	A1
A2A1B1C1C2D2D1	18	1	19	B1
A2A1B1C1C2D2E2	12	0	12	C1
				C2
				B2
				D2

Solution: A2A1B1C1C2D2E2

Cost: 12



Tour de AI with A*

Node	g	h	f	
A2A1B1B2	12	3	15	Closed set
A2A1B1C1D1	14	1	15	A2
A2A1B1C1C2D2D1	18	1	19	A1
A2A1B1C1C2D2E2	12	0	12	B1
				C1
				C2
				B2
				D2

Nodes expanded:

A2

A2A1

A2A1B1

A2A1B1C1

A2A1B1C1C2

A2B2

A2A1B1C1C2D2

Solution: A2A1B1C1C2D2E2

Cost: 12

iClicker: Tour de AI with UCS?

What is the cost of the UCS solution?

	1	2
A	4	4 
B	3	3 
C	2	2
D	1 	1
E	 Finish Line 	

Tour de AI with UCS?

Solution?

A2A1B1C1C2D2E2

	1	2
A	4	4 
B	3	3 
C	2	2
D	1 	1
E	 Finish Line 	

CSPs: Question Examples

- ▶ Domain after enforcing node consistency
- ▶ Domain after enforcing arc consistency
- ▶ Domain after running the AC-3 algorithm to completion
- ▶ Ordering (MRV, Least Constraining value)
- ▶ Structure: tree structured CSP, cutset conditioning, smallest cutset

Review: CSPs

- ▶ Initial domains: {1, 2, 3, 4, 5, 6}
- ▶ Constraints:
 - A is even
 - $A + B \leq 5$
- ▶ What are the domains of A and B after enforcing **node consistency**?



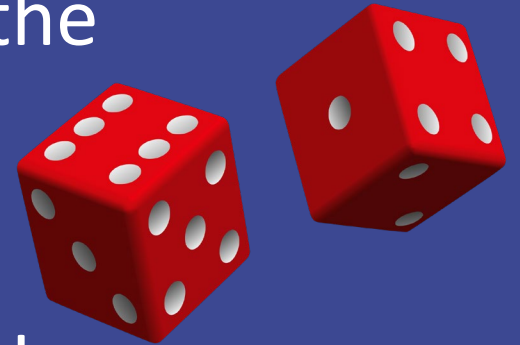
Review: CSPs

- ▶ Constraints:
 - A is even (unary)
 - $A + B \leq 5$ (binary)
- ▶ What are the domains of A and B after enforcing **node consistency**?
 - $A = \{2, 4, 6\}$
 - $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing the consistency of arc $A \rightarrow B$?



Review: CSPs

- ▶ Constraints: A is even (unary), $A + B \leq 5$ (binary)
- ▶ What are the domains of A and B after enforcing **node consistency**? $A = \{2, 4, 6\}$ $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing the consistency of arc $A \rightarrow B$?
 - $A = \{2, 4\}$ $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing the consistency of arc $B \rightarrow A$?



Example

- ▶ Constraints: A is even (unary), $A + B \leq 5$ (binary)
- ▶ What are the domains of A and B after enforcing **node consistency**? $A = \{2, 4, 6\}$ $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing **arc consistency** of arc $A \rightarrow B$?
 - $A = \{2, 4\}$ $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing **arc consistency** of arc $B \rightarrow A$?
 - $A = \{2, 4\}$ $B = \{1, 2, 3\}$

iClicker:

What is the domain of A after enforcing the consistency of the whole CSP - running the AC-3 algorithm to completion?

Example

- ▶ Constraints: A is even (unary), $A + B \leq 5$ (binary)
- ▶ What are the domains of A and B after enforcing **node consistency**? $A = \{2, 4, 6\}$ $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing the consistency of arc $A \rightarrow B$?
 - $A = \{2, 4\}$ $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing the consistency of arc $B \rightarrow A$?
 - $A = \{2, 4\}$ $B = \{1, 2, 3\}$



iClicker:
How many solutions
are there?

Example

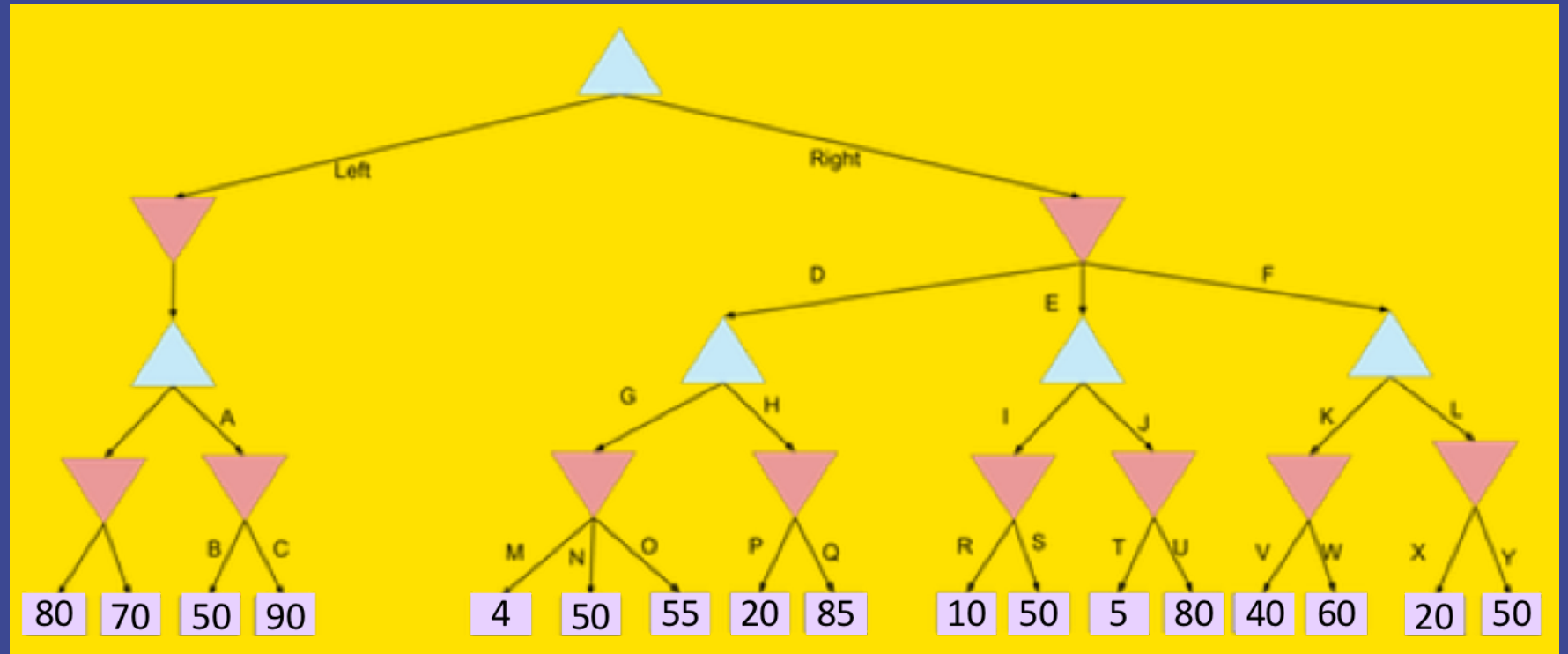
- ▶ Constraints: A is even (unary), $A + B \leq 5$ (binary)
- ▶ What are the domains of A and B after enforcing **node consistency**? $A = \{2, 4, 6\}$ $B = \{1, 2, 3, 4, 5, 6\}$
- ▶ What are the domains of A and B after enforcing the consistency of arc $A \rightarrow B$?
 - $A = \{2, 4\}$ $B = \{1, 2, 3, 4, 5\}$
- ▶ What are the domains of A and B after enforcing the consistency of arc $B \rightarrow A$?
 - $A = \{2, 4\}$ $B = \{1, 2, 3\}$

iClicker:

What is one possible assignment returned by the backtracking algorithm? Please write the assignment as a dictionary - for example: `{'A': 1, 'B': 2}`

Adversarial Search

- What is the minimax value at the root?

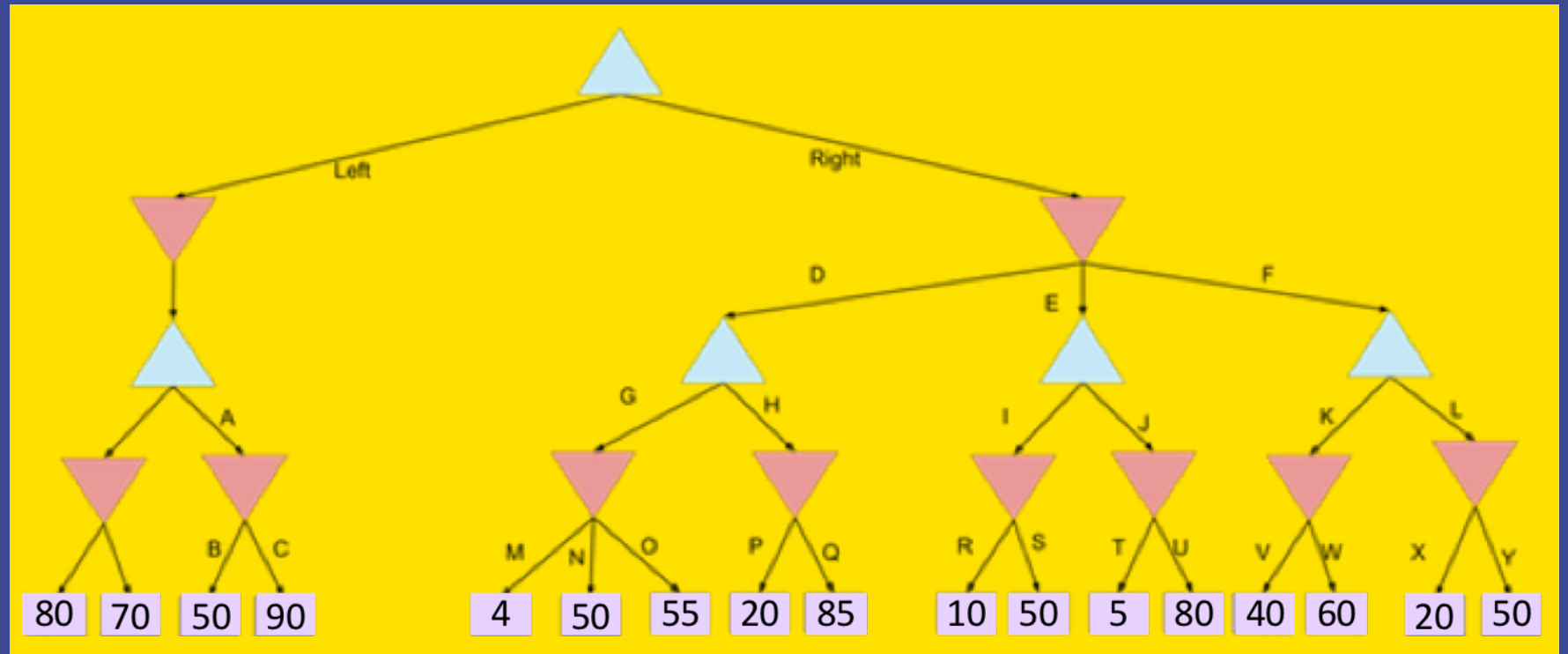


Adversarial Search

► Which action should the maximizing player (MAX) take?

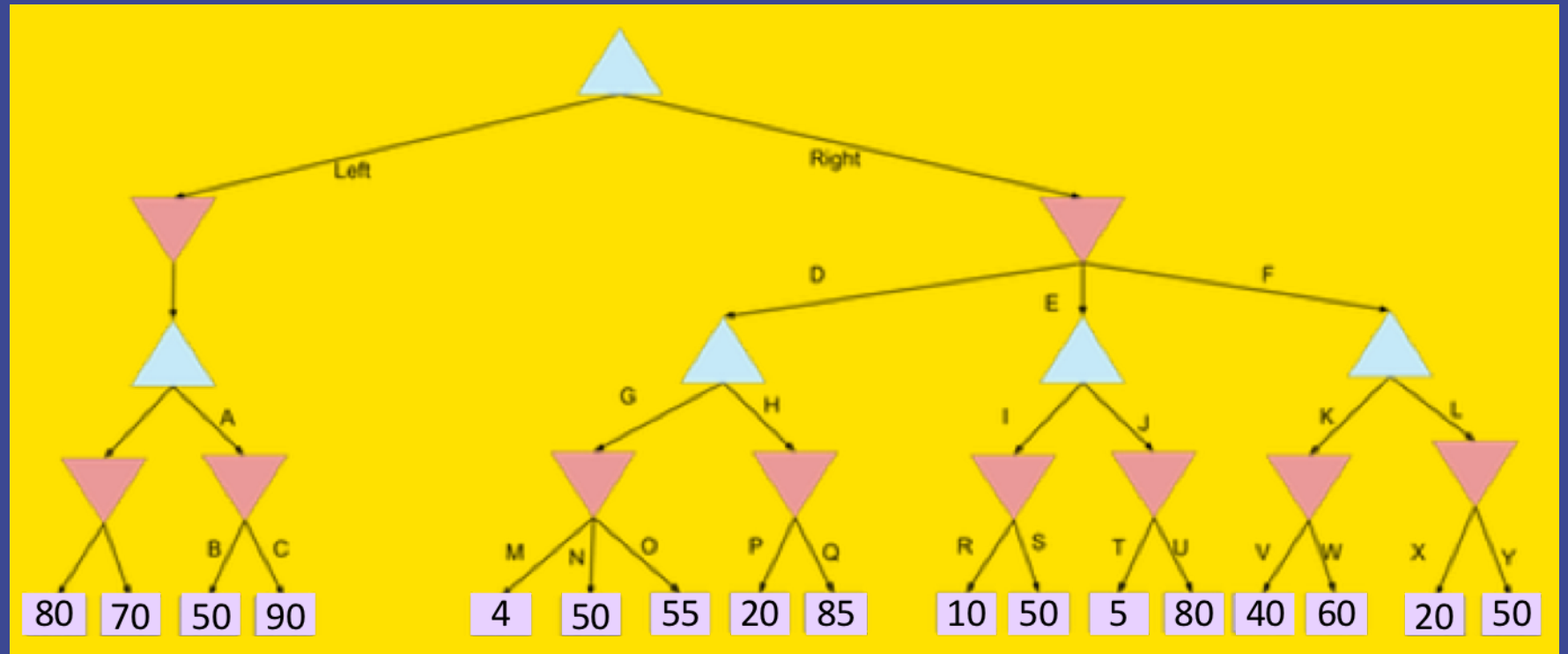
A. Left

B. Right



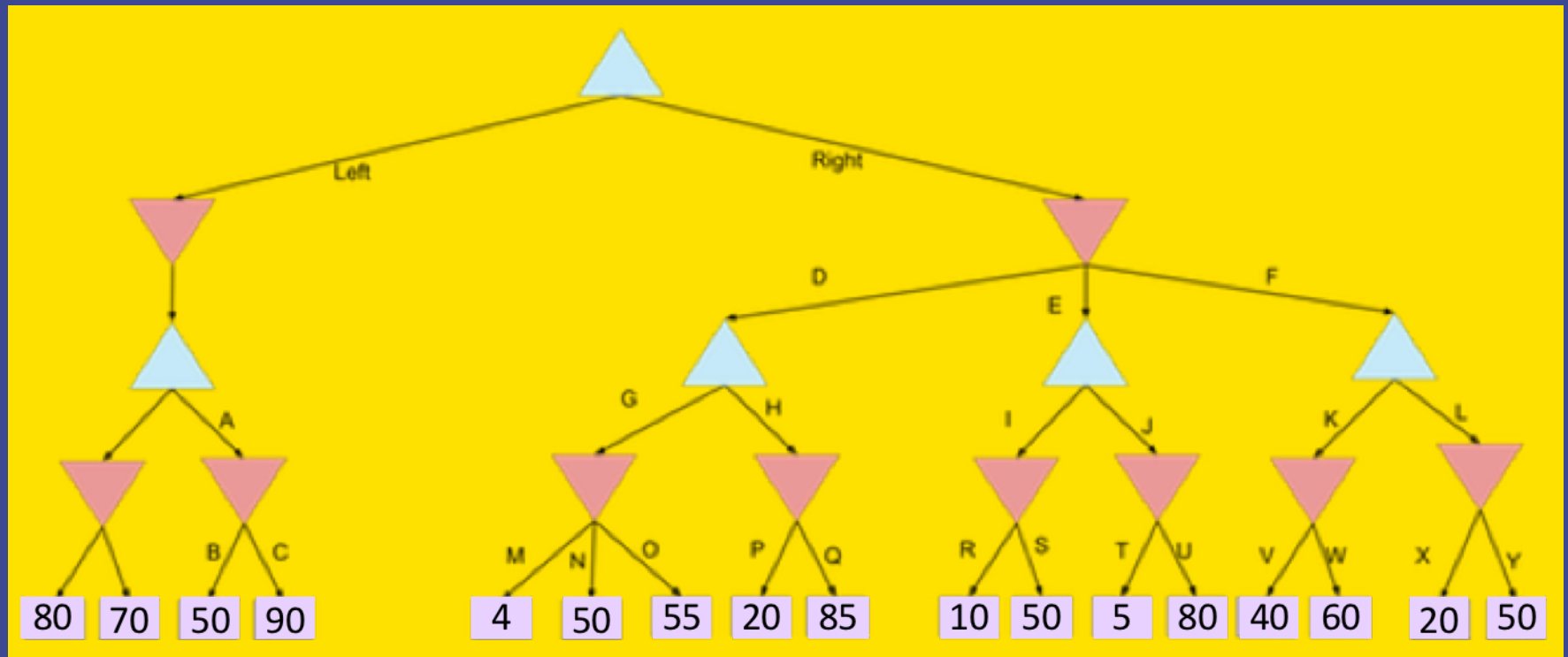
Adversarial Search

- ▶ Which branches will be pruned when we apply alpha-beta pruning? If there is more than one branch, please enter them separated by a comma.



Adversarial Search

- ▶ Which branches will be pruned when we apply alpha-beta pruning? If there is more than one branch, please enter them separated by a comma.
- ▶ C, N, O, Q, E, F



Reminders

- ▶ Homework 4: due October 16
- ▶ Midterm on Thursday
- ▶ Good luck Everyone!